ETSI TS 103 833-2 V1.1.1 (2022-12)

Core Network and Interoperability Testing (INT);
Conformance Test Specifications for the SCC-AS Services;
(3GPP™ Release 16);
Part 2: Test Suite Structure (TSS) and Test Purposes (TP)

Reference
DTS/INT-00182-2

Keywords
TSS&TP

ETSI

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Core Network and Interoperability Testing (INT).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [3].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides the Test Suite Structure (TSS) and Test Purposes (TP for conformance test specifications for the SCC-AS Services as specified in ETSI TS 124 237 [1] and ETSI TS 124 292 [2] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [5] and ETSI ETS 300 406 [6].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 124 237: "Universal Mobile Telecommunications System (UMTS); LTE; IP Multimedia (IM) Core Network (CN) subsystem IP Multimedia Subsystem (IMS) service continuity; Stage 3 (3GPP TS 24.237 Release 16)".

[2] ETSI TS 124 292: "Universal Mobile Telecommunications System (UMTS); LTE; IP Multimedia (IM) Core Network (CN) subsystem Centralized Services (ICS); Stage 3 (3GPP TS 24.292 Release 16)".

[3] ETSI TS 103 833-1: "Core Network and Interoperability Testing (INT); Conformance Test Specifications for the SCC-AS Services; (3GPPTM Release 16); Part 1: Protocol Implementation Conformance Statement (PICS)".

[4] ISO/IEC 9646-1: "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 1: General concepts".

[5] ISO/IEC 9646-7: "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 7: Implementation Conformance Statements".

[6] ETSI ETS 300 406: "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

[7] ETSI ES 203 119-4: "Methods for Testing and Specification (MTS); The Test Description Language (TDL); Part 4: Structured Test Objective Specification (Extension)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] IETF RFC 6809: "Mechanism to Indicate Support of Features and Capabilities in the Session Initiation Protocol (SIP)".

[i.2] IETF RFC 6086: "Session Initiation Protocol (SIP) INFO Method and Package Framework".

[i.3] IETF RFC 3891: "The Session Initiation Protocol (SIP) "Replaces" Header".

[i.4] IETF RFC 4488: "Suppression of Session Initiation Protocol (SIP) REFER Method Implicit Subscription".

[i.5] IETF RFC 4538: "Request Authorization through Dialog Identification in the Session Initiation Protocol (SIP)".

[i.6] IETF RFC 3261: "SIP: Session Initiation Protocol".

[i.7] ETSI TS 124 229: "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3 (3GPP TS 24.229)".

[i.8] ETSI TS 129 292: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Interworking between the IP Multimedia (IM) Core Network (CN) subsystem (IMS) and MSC Server for IMS Centralized Services (ICS) (3GPP TS 29.292)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TS 124 237 [1] and ETSI TS 124 292 [2] apply.

Abstract Test Method (ATM): Refer to ISO/IEC 9646-1 [4].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [4].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [4].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [4].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TS 124 237 [1], ETSI TS 124 292 [2] and the following apply:

ATCF	Access Transfer Control Function
I-CSCF	Interrogating CSCF
MSC	Mobile Switching Center
SCC AS	Service Centralization and Continuity Application Server
S-CSCF	Serving CSCF
TP	Test Purpose
TSS	Test Suite Structure
UE	User Equipment

4 Test configurations

4.1 Introduction

Test purposes of the present document address the IMS functional entities that are accessible via the following standardized interfaces:

- ISC; and
- Gm.

4.2 Test configurations

The ISC interface is located between an IMS with UE and the SCC-AS.

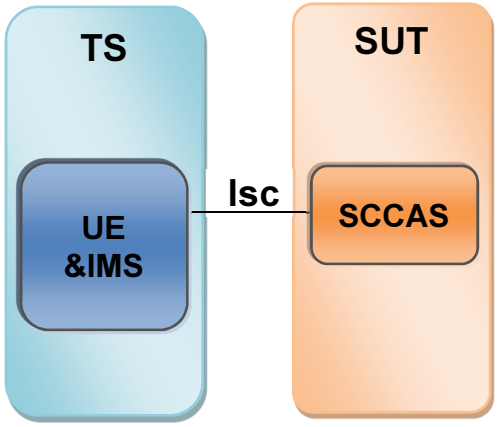

Figure 1: Test configuration CF_SCCAS_01

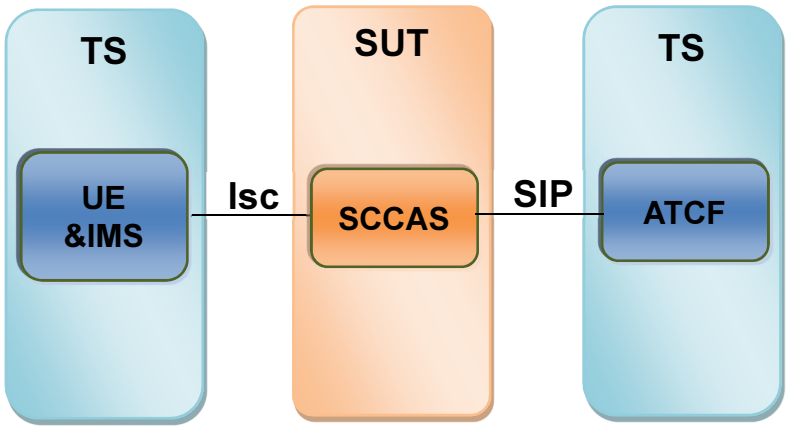

Figure 2: Test configuration CF_SCCAS_02

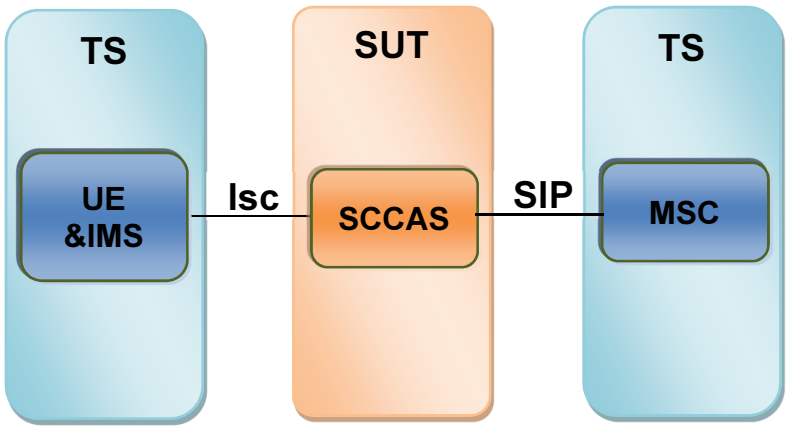

Figure 3: Test configuration CF_SCCAS_03

5 Test Suite Structure (TSS) and Test Purposes (TP)

5.1 Test Suite Structure

5.1.1 TP naming convention

TPs are numbered, starting at 01, within each group. Groups are organized according to the TSS.

Table 1: TP identifier naming convention scheme

Identifier: **<TP>_<interface>_<iut>_<scope> _<method> _<nn>**				
<tp>	=	Test Purpose:	fixed to "TP"	
<interface>		Interface:	ISC	
<iut>	=	type of IUT:	SCCAS	
<scope>	=	group	RSC	Registration in the IM CN subsystem for service continuity
			GEN	General Capabilities
			ORI	Call origination for service continuity
			TER	Call termination for service continuity
			CPT	CS-PS access transfer
			PCT	PS-CS access transfer
			PPT	PS-PS access transfer
			CON	PS-PS access transfer in conjunction with PS-CS access transfer
			SRA	PS-CS access transfer, Single Radio
			MED	Media adding/deleting for access transfer
<method>	=	subgroup	REG	Register
			INV	Invite
			RIN	ReInvite
			REF	Refer
			INF	Info
			UPD	Update
			BYE	Bye
			CAN	Cancel
			RES	Response
<nn>	=	sequential number	(01 to 99)	

5.1.2 Test strategy

As the base standards ETSI TS 124 237 [1] and ETSI TS 124 292 [2] contain no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification ETSI TS 103 833-1 [3].

5.1.3 TP structure

Each TP has been written in TDL-TO and thus in a structured manner which is consistent with all other TPs. The intention of this is to make the TPs more formal. In addition, a more readable format is provided by generating tables out of the TDL-TO format. The defined structure, that has been used, is illustrated in table 2. This table should be read in conjunction with any TP, i.e. please use a TP as an example to facilitate the full comprehension of table 2. All structures are defined formally in the TDL Specification ETSI ES 203 119-4 [7]. The TDL-TO files are also included as an electronic annex to the present document

Table 2: Structure of a single TP

TP part	Text	Example
Header	<Identifier> <Test objective> <Reference> <PICS reference>	*See table 2* *"The IUT is responding on a correctly set …"* *ETSI TS 124 237#section-3* *PIC_Server*
Initial condition (optional)	Free text description of the condition that the IUT has reached before the test purpose applies.	*… the IUT is in the initial state …*
Start point	Describes the full logic of the test purpose. Includes trigger and expected behaviour of the IUT.	*Expected behaviour* *ensure that { … }*
Trigger	One or more actions that trigger an expected response of the IUT. Mostly a set of different messages the IUT receives.	*when {* *the IUT entity receives an INVITE request message* *containing CSeq indicating value 1 …* *}*
Expected behaviour	Describes the response that the IUT sends after receiving a certain (set of) messages. This response describes the pass criteria	*then {* *the IUT entity sends a 100 Trying response message* *containing CSeq indicating value 1 …* *}*

5.2 Test Purposes

5.2.1 PICS references

All PICS items referred to in this clause are as specified in ETSI TS 103 833-1 [3] unless indicated otherwise by another numbered reference. PICS items are only meant for test selection, therefore only PICS items with status optional or conditional are explicitly mentioned.

5.2.2 SCC AS Role

5.2.2.1 Test selection

The IUT takes the role of the SCC AS; PICS A.2/x and applicable test configuration is CF_SCCAS_01 if not specified differently in the TP.

5.2.2.2 Registration in the IM CN subsystem for service continuity

TP Id	TP_ISC_SCCAS_RSC_REG_01
Test Objective	Verify that the SCCAS successfully obtain/processes 3pty registration information.
Reference	ETSI TS 124 237 [1], Clause 6.3.1
PICS Selection	NONE
Initial Conditions	

with {
the UE isNotRegisteredTo the IMS_A
}	

Expected Behaviour	
ensure that { when {	

```
        the IUT receives a REGISTER containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            SipMessageBody containing
                MIME indicating value "REGISTER from_UE_A",
                MIME indicating value "r_200_Ok to_UE_A"
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI //,
            //Path indicating value PX_P_CSCF_A_SIP_URI,
            //PChargingVector containing
            //    icid indicating value PX_TO_BE_DEFINED,
            //PVisitedNetworkID indicating value PX_TO_BE_DEFINED,
            //Require indicating value "path",
            //Supported indicating value "path"

        to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_RSC_REG_02
Test Objective	Verify that the SCCAS successfully obtain/processes de-registration information (Expires=0).
Reference	ETSI TS 124 237 [1], Clause 6.3.1
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REGISTER containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Expires indicating value 0
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA
        to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_RSC_REG_03
Test Objective	Verify that the SCCAS successfully obtain/processes SRVCC information.
Reference	ETSI TS 124 237 [1], Clause 6.3.2 (1st numbered list)
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isNotRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REGISTER containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
```

```
            FeatureCaps containing
                g3gppAtcfFeatureCapabilityIndicator,
            SipMessageBody containing
                MIME indicating value "REGISTER from_UE_A",
                MIME indicating value "r_200_Ok to_UE_A" containing Contact
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI
        to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_RSC_REG_04
Test Objective	Verify that the SCCAS successfully sends MESSAGE providing PS to CS SRVCC information.
Reference	ETSI TS 124 237 [1], Clause 6.3.3
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC_ATCF with {
the UE isNotRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REGISTER containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            FeatureCaps containing
                g3gppAtcfFeatureCapabilityIndicator indicating value "g.3gpp.atcf-mgmt-uri",
            SipMessageBody containing
                MIME indicating value "REGISTER from_UE_A",
                MIME indicating value "r_200_Ok to_UE_A" containing Contact
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI //,
            //Path indicating value PX_P_CSCF_A_SIP_URI,
            //PChargingVector containing
            //    icid indicating value PX_TO_BE_DEFINED,
            //PVisitedNetworkID indicating value PX_TO_BE_DEFINED,
            //Require indicating value "path",
            //Supported indicating value "path"

        to the IMS_S_CSCF entity
        and the IUT sends a MESSAGE containing
         RUri indicating value "ATCFManagementUri",
         PAsertedIdentity,
         SipMessageBody containing
                MIME indicating value "application/vnd.3gpp.SRVCC-info+xml",
         PChargingVector indicating value "type 1 orig-ioi"
        to the ATCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_RSC_REG_05
Test Objective	Verify that the SCCAS successfully get CS to PS SRVCC information.
Reference	ETSI TS 124 237 [1], Clause 6.3.4
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC_MSC with {
the UE isNotRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REGISTER containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact containing
               g3gppCs2psSsrvccTag,
               g3gppPathTag
            FeatureCaps
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI

        to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_RSC_REG_06
Test Objective	Verify that the SCCAS successfully provide CS to PS SRVCC information.
Reference	ETSI TS 124 237 [1], Clause 6.3.5
PICS Selection	NONE

Initial Conditions

```
//CS2PSSRVCC with {
the UE isNotRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REGISTER containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact containing
               SipInstanceTag,
               g3gppIcsiRefTag
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI,
            AcceptContact containing
               g3gppPathMediaFeatureTag
            //Path indicating value PX_P_CSCF_A_SIP_URI,
            PChargingVector indicating value "type 1 orig-ioi",
            //PVisitedNetworkID indicating value PX_TO_BE_DEFINED,
            PassertedIdentity indicating value PX_SCCAS_URI
            //Supported indicating value "path"
            SipMessageBody containing
                MIME indicating value "application/vnd.3gpp.srvcc-ext+xml"

        to the IMS_S_CSCF entity
```

```
    }
}
```

5.2.2.3 General capabilities

TP Id	TP_ISC_SCCAS_GEN_INV_01
Test Objective	Verify that the SCCAS send INVITE for anchored user populated with FeatureCaps header.
Reference	ETSI TS 124 237 [1], Clauses 6A.4.2 and 4.2.2
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS_A
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a INVITE
from the IMS_S_CSCF entity
}
then {
the IUT sends a INVITE containing
From indicating value PX_IMS_S_CSCF_SIP_URI,
To indicating value PX_UE_A_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
//...
FeatureCaps containing //according to RFC6809
g3gppFeatureCapabilityIndicator, //see Annex C
g3gppRemoteLegInfoCapabilityIndicator, //see Annex C
Accept containing
MIME indicating value "application/vnd.3gpp.state-and-event-info.xml", //subclause D.2.3
RecvInfo containing
g3gppStateAndEventPackageName //according to RFC6086
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_GEN_INV_02
Test Objective	Verify that the SCCAS sends 1xx and/or 2xx response to the SIP INVITE request towards the served user.
Reference	ETSI TS 124 237 [1], Clauses 6A.4.3 and 4.2.2
PICS Selection	NONE
Initial Conditions	

//SCCAS_PS2CS_SRVCC, SCCAS_PS2CS_SRVCC_ALERTING,SCCAS_PS2CS_SRVCC_PRE-ALERTING with {
the UE isRegisteredTo the IMS_A and
the UE hasInitiatedDialogWith the UE_TERM
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a r_183_SessionProgress containing
From indicating value PX_IMS_S_CSCF_SIP_URI,
To indicating value PX_UE_A_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
//Contact containing
// g3gppMidcallMediaFeature Tag, //list -B
// g3gppSrvccAlertingFeature tag, //list -C
// g3gppPs2csSrvccOrigPreAlertingMediaFeature Tag, //list -D
FeatureCaps,
SipMessageBody
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_183_SessionProgress containing
From indicating value PX_IMS_S_CSCF_SIP_URI,
To indicating value PX_UE_A_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
//...
FeatureCaps containing //IETF RFC6809	

```
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaIndicator, //list -D
                g3gppRemoteLegInfoFeatureCapabilityIndicator //list -E
            //Contact containing
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_GEN_INV_03
Test Objective	Verify that the SCCAS sends 1xx and/or 2xx response to the SIP INVITE request due to STN-SR or STN.
Reference	ETSI TS 124 237 [1], Clauses 6A.4.3A (1st Numbered list, item 1) and 4.2.2
PICS Selection	NONE
Initial Conditions	

```
//SCCAS_PS2CS_SRVCC, SCCAS_PS2CS_SRVCC_ALERTING,SCCAS_PS2CS_SRVCC_PRE-ALERTING with {
the UE isRegisteredTo the IMS_A  and
the UE hasInitiatedDialogWith the UE_TERM
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_STNSR_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact containing
                g3gppMidcallMediaFeature_Tag,
                g3gppSrvccAlertingFeature_Tag,
                g3gppPs2csSrvccOrigPreAlertingMediaFeature_Tag
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_STNSR_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //...
            Contact indicating value UE_CONTACT,
            PAssertedIdentity indicating value UE_PAI,
            FeatureCaps containing //according to RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppRemoteLegInfoCapabilityIndicator //see Annex C
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_GEN_RIN_01
Test Objective	Verify that the SCCAS sends 1xx and/or 2xx response to the SIP INVITE target refresh request towards the served user.
Reference	ETSI TS 124 237 [1], Clauses 6A.4.5 and 4.2.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS_A and
```

```
the UE hasEstablishedDialog the SCCAS
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact containing
                g3gppMidcallMediaFeature Tag, //list -B
                g3gppSrvccAlertingFeature tag, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaFeature Tag, //list -D
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then{
        the IUT sends a INVITE
        to the IMS_S_CSCF and
        the IUT receives a r_183_SessionProgress
        from the IMS_S_CSCF and
        the IUT sends a r_183_SessionProgress containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //...
            FeatureCaps containing //IETF RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaIndicator, //list -D
            //Contact containing
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_GEN_RIN_02
Test Objective	Verify that the SCCAS sends the SIP INVITE target refresh request towards the served user.
Reference	ETSI TS 124 237 [1], Clauses 6A.4.5 and 4.2.2
PICS Selection	NONE

Initial Conditions

```
with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact containing
                g3gppMidcallMediaFeature Tag, //list -B
                g3gppSrvccAlertingFeature tag, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaFeature Tag, //list -D
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
```

```
            Via indicating value PX_UE_A_VIA,
            //...
            FeatureCaps containing //IETF RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
            //Contact containing
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_GEN_REF_01
Test Objective	Verify that the SCCAS sends 403 response to the SIP REFER request from remote UE.
Reference	ETSI TS 124 237 [1], Clause 6A.4.6 (1st Numbered list)
PICS Selection	NONE
Initial Conditions	

```
//PICS_SCCAS_REJECTION_OF_MALICIOUS_REFER with {
the UE isRegisteredTo the IMS_A and
the UE hasInitiatedDialogWith the SCCAS
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a REFER containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //Contact
            ReferSub indicating value "false",
            Supported indicating value "norefersub",
            ReferTo  containing "a SIP URI with the Target-Dialog_URI_header_field",
            SipMessageBody containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3

        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_403_Forbidden containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA
            //...

        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_GEN_INF_01
Test Objective	Verify that the SCCAS not include Accept,RecvInfo towards remote UE.
Reference	ETSI TS 124 237 [1], Clause 6A.4.7
PICS Selection	NONE
Initial Conditions	

```
//SCCAS_PS2CS_SRVCC, SCCAS_PS2CS_SRVCC_ALERTING with {
the UE isRegisteredTo the IMS_A and
the UE hasInitiatedDialogWith the SCCAS
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
```

```
            Contact containing
                g3gppMidcallMediaFeature Tag, //list -B
                g3gppSrvccAlertingFeature tag, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaFeature Tag, //list -D
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //...
            FeatureCaps containing //IETF RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
            //Contact
            Accept not indicating value "application/vnd.3gpp.state-and-event-info.xml",
//subclause D.2.3
            RecvInfo not indicating value g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF entity
     }
}
```

5.2.2.4 Call origination for service continuity

TP Id	TP_ISC_SCCAS_ORI_INV_01
Test Objective	Verify that the SCCAS first receives SIP INVITE request from UE/SCSCF.
Reference	ETSI TS 124 237 [1], Clauses 7.3.1 and 7.3.2 ETSI TS 124 292 [2], Clause 7.4.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value "sccas as topmost entry",
            RecordRoute indicating value PX_UE_A_ACCEPT,
            PAssertedIdentity indicating value PX_UE_A_PAS,
            Accept indicating value PX_UE_A_ACCEPT,
            Contact indicating value PX_UE_A_CONTACT,
            AcceptContact indicating value PX_UE_A_ACCEPT_CONTACT,
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_ORI_INV_02
Test Objective	Verify that the SCCAS sends 1xx/2xx response to the SIP INVITE request towards the served user.
Reference	ETSI TS 124 237 [1], Clauses 7.3.2, 6A.4.3 and 4.2.2 ETSI TS 124 292 [2], Clause 7.4.2
PICS Selection	NONE

Initial Conditions

```
//SCCAS_PS2CS_SRVCC, SCCAS_PS2CS_SRVCC_ALERTING,SCCAS_PS2CS_SRVCC_PRE-ALERTING with {
the UE isRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value "sccas as topmost entry",
            RecordRoute indicating value PX_UE_A_ACCEPT,
            PAssertedIdentity indicating value PX_UE_A_PAS,
            Accept indicating value PX_UE_A_ACCEPT,
            AcceptContact indicating value PX_UE_A_ACCEPT_CONTACT,
            Contact containing
                g3gppMidcallMediaFeature Tag, //list -B
                g3gppSrvccAlertingFeature tag, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaFeature Tag, //list -D
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a INVITE
        to the IMS_S_CSCF and
        the IUT receives a r_183_SessionProgress
        from the IMS_S_CSCF and
        the IUT sends a r_183_SessionProgress containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //...
            FeatureCaps containing //IETF RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaIndicator, //list -D
            //Contact containing
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086

        to the IMS_S_CSCF and
        the IUT receives a r_200_Ok
        from the IMS_S_CSCF and
        the IUT sends a r_200_Ok containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_A_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            //...
            FeatureCaps containing //IETF RFC6809
                g3gppFeatureCapabilityIndicator, //see Annex C
                g3gppMidcallFeatureCapabilityIndicator, //list - B
                g3gppSrvccAlertingFeatureCapabilityIndicator, //list -C
                g3gppPs2csSrvccOrigPreAlertingMediaIndicator, //list -D
                g3gppRemoteLegInfoFeatureCapabilityIndicator //list -E
            Contact,
            Supported indicating value "tdialog,replaces"
            Accept containing
                MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
            RecvInfo containing
                g3gppStateAndEventPackageName //according to RFC6086
```

```
        to the IMS_S_CSCF entity
     }
}
```

5.2.2.5 Call termination for service continuity

TP Id	TP_ISC_SCCAS_TER_INV_01
Test Objective	Verify that the SCCAS last receives SIP INVITE request from UE/SCSCF.
Reference	ETSI TS 124 237 [1], Clauses 8.3.1 and 8.3.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS_A and
the UE_TERM isRegisteredTo the IMS_A
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            RequestUri indicating value PX_UE_TERM_SIP_URI,
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_TERM_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value "sccas as topmost entry",
            PServedUser indicating value PX_UE_TERM_SIP_URI,
            Contact,
            FeatureCaps,
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a INVITE containing
            From indicating value PX_IMS_S_CSCF_SIP_URI,
            To indicating value PX_UE_TERM_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA
        to the IMS_S_CSCF entity
     }
}
```

5.2.2.6 PS-CS access transfer

TP Id	TP_ISC_SCCAS_PCT_INV_01
Test Objective	Verify that the SCCAS distinguish SIP INVITE with STI for subscribed UE due to PS to CS.
Reference	ETSI TS 124 237 [1], Clause 9.3.1
PICS Selection	NONE
Initial Conditions	

```
//SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            RequestUri indicating value PX_STI_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STI_VIA,
            Contact,
            TargetDialog containing "dialog info from PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a ReINVITE
        to the IMS_S_CSCF entity
```

```
    }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_01
Test Objective	Verify that the P-CSCF successfully processes an initial INVITE (Originating Leg) with dynamic STI.
Reference	ETSI TS 124 237 [1], Clauses 9.3.3 and 10.3.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,        // this indicates dynamic STI
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            Supported indicating value "replaces",
            PAccessNetworkInfo,
            Replaces containing  // Replaces indicating value RFC 3891
                 call_id indicating value PX_call_id,
                 from_tag indicating value PX_from_tag,
                 to_tag indicating value PX_to_tag
            MessageBody containing
                SDP containing  // SDP: Session Description Protocol
                    Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an r_200_Ok containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            PAccessNetworkInfo,
            ContentType indicating value "application/sdp",
            ContentLength,
            MessageBody containing
                SDP containing
                     Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_CPT_INV_02
Test Objective	Verify that the P-CSCF successfully processes an initial INVITE (Originating Leg) with static STI.
Reference	ETSI TS 124 237 [1], Clause 9.3.3
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
            From indicating value PX_UE_A_TEL_URI,        // this indicates static STI
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            Supported indicating value "replaces",
            PAccessNetworkInfo,
            Replaces containing  // Replaces indicating value RFC 3891
                 call_id indicating value PX_call_id,
                 from_tag indicating value PX_from_tag,
```

```
                to_tag indicating value PX_to_tag
            MessageBody containing
                SDP containing  // SDP: Session Description Protocol
                    Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an r_200_Ok containing
            From indicating value PX_UE_A_TEL_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            PAccessNetworkInfo,
            ContentType indicating value "application/sdp",
            ContentLength,
            MessageBody containing
                SDP containing
                     Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_CPT_RIN_01
Test Objective	Verify that the P-CSCF successfully processes an re-INVITE (Originating Leg).
Reference	ETSI TS 124 237 [1], Clause 9.3.3
PICS Selection	NONE

Initial Conditions

```
with {
the UE isRegisteredTo the IMS and
the UE previouslyEstablishedCallWith
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing      // re-invite
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            Supported indicating value "replaces",
            PAccessNetworkInfo,
            Replaces containing  // Replaces indicating value RFC 3891
                 call_id indicating value PX_call_id,
                 from_tag indicating value PX_from_tag,
                 to_tag indicating value PX_to_tag
            MessageBody containing
                SDP containing  // SDP: Session Description Protocol
                    Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an r_200_Ok containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            PAccessNetworkInfo,
            ContentType indicating value "application/sdp",
            ContentLength,
            MessageBody containing
                SDP containing
                     Media indicating value PX_PSMediaOnly,
                    Version indicating value "0"
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_CPT_ACK_01
Test Objective	Verify that the P-CSCF successfully processes an ACK and release the source access leg.
Reference	ETSI TS 124 237 [1], Clauses 9.3.3 and 9.3.6.
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the UE hasReceived200OkOnInitialRequestForDialogWith
}
Expected Behaviour	

ensure that {
when {
the IUT receives an ACK containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE
from the IMS_S_CSCF entity
}
then {
the IUT sends an BYE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CPT_INV_03
Test Objective	Verify that the SCC AS Conditions for selecting a sessions in an early dialog phase.
Reference	ETSI TS 124 237 [1], Clauses 9.3.4 and 6A.4.3
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
// TODO: SCC AS supports the MSC Server assisted mid-call feature according to operator policy
}
Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
Contact containing
g3gppSrvccAlertingFeatureCapabilityIndicator Tag
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_2xx_Any containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
FeatureCaps containing
g3gppMidcallFeatureCapabilityIndicator,
MessageBody containing
SDP containing
Media indicating value PX_PSMediaOnly
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CPT_INV_04
Test Objective	Verify that the SCC AS accepts conditions in an early dialog phase.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.1
PICS Selection	NONE
Initial Conditions	

```
// PICS_SCCAS_PS2CS_DUAL_RADIO_ACCES_TRANSFER_OF_CALLS_IN_EARLY_DIALOG with {
the UE isRegisteredTo the IMS
// TODO: SCC AS supports the MSC Server g.3gpp.drvcc-alerting feature
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        Contact containing
            g3gppDrvccAlertingFeature tag //Annex C.15
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an r_2xx_Any containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        FeatureCaps containing
            g3gppDrvccAlertingFeatureCapabilityIndicator
    to the IMS_S_CSCF entity
    }
                  }
```

TP Id	TP_ISC_SCCAS_PCT_UPD_01
Test Objective	Verify that the SCC AS sends a SIP UPDATE after receiving a SIP INVITE request due to PS to CS STN and if there are one or more dialogs in an early dialog phase and the remote leg is not a precondition enabled dialog.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.2
PICS Selection	NONE
Initial Conditions	

```
// PICS_SCCAS_PS2CS_DUAL_RADIO_ACCES_TRANSFER_OF_CALLS_IN_EARLY_DIALOG with {
the UE isRegisteredTo the IMS
// TODO:
// here are one or more dialogs in an early dialog phase supporting a session with active speech
media component such that:
//        1 all dialogs are early dialogs created by the same SIP INVITE request
//        2 a SIP 180 Ringing response to SIP INVITE request was received in at least one of those
early dialogs
//        3 a g.3gpp.drvcc-alerting feature-capability indicator as described in annex C was
included in a Feature-Caps
//        header field by the SCC AS in the SIP 180 Ringing response and
//        4 the Contact header field in the initial SIP INVITE request sent by the SC UE towards
the SCC AS included the
//        g.3gpp.drvcc-alerting media feature tag as described in annex C,
// and
// the remote leg is not a precondition enabled dialog
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an UPDATE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
```

```
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        // 1 set the Request-URI set to the URI contained in the Contact header field returned at
the creation of the dialog
        // with the remote UE
        Contact indicating value "set to the Contact header field provided by the served UE at the
creation of the dialog",// 2 the Contact header field set to the Contact header field provided by
the served UE at the creation of the dialog with the remote UE
        ContentType indicating value "application/sdp",
            MessageBody containing  // 3 a new SDP offer, including:
                SDP containing
                    Media indicating value "the media characteristics as received in the SIP
INVITE request due to PS to CS STN including any precondition mechanism specific SDP attributes"


    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_PCT_RES_02
Test Objective	Verify that the SCC AS sends SIP provisional response to the SIP INVITE request due to PS to CS STN.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.2
PICS Selection	NONE
Initial Conditions	

```
// TODO with {
the UE isRegisteredTo the IMS
// a SIP INVITE request due to PS to CS STN tooks place
// and
// SCC AS has send a SIP UPDATE request according to TP TP_ISC_SCCAS_PCT_UPD_01
// and
// the remote leg is not a precondition enabled dialog
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an r_200_Ok containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an r_1xx_Any containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        PEarlyMedia indicating value "the last received P-Early-Media header field",
        ContentType indicating value "application/sdp",
            MessageBody containing  // 3 a new SDP offer, including:
                SDP containing
                    Media indicating value "SDP answer based on the SDP answer received from the
remote UE"


    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_PCT_INV_05
Test Objective	Verify that the SCC AS send a SIP UPDATE request towards the remote UE after receiving a SIP INVITE request due to PS to CS STN in early dialog phase supporting a session with active speech media component such.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.3
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
```

```
// one dialog is early dialog
// and
// a SIP 180 Ringing response to the SIP INVITE request has not been received yet in any of the
existing dialogs
// and
// the SCC AS included a g.3gpp.ps2cs-drvcc-orig-pre-alerting feature-capability indicator as in a
Feature-Caps header field of SIP 18x responses
// and
// the Contact header field in the initial SIP INVITE request sent by the SC UE towards the SCC AS
included a g.3gpp.ps2cs-drvcc-orig-pre-alerting media feature tag as
// and
// the remote leg is not a precondition enabled dialog
               }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an UPDATE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        PEarlyMedia indicating value "the last received P-Early-Media header field",
        ContentType indicating value "application/sdp",
            MessageBody containing  // 3 a new SDP offer, including:
                SDP containing
                    Media indicating value "the media characteristics as received in the SIP
INVITE request due to PS to CS STN but excluding any precondition mechanism specific SDP
attributes"


    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_PCT_REF_01
Test Objective	Verify that the SCC AS populate a SIP REFER request to transfer an additional session in PS to CS dual radio access.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.4
PICS Selection	NONE

Initial Conditions

```
// TODO with {
the UE isRegisteredTo the IMS
// TODO: initial condition SIP INVITE request to PS to CS STN
// "In order to transfer an additional session on the originating side that can be in pre-alerting
phase or in an alerting phase,
// the SCC AS shall send a SIP REFER request in the
// dialog created by the SIP INVITE request due to PS to CS STN"
           }
```

Expected Behaviour

```
// 1. the Refer-Sub header field with value "false" as specified in IETF RFC 4488
            // 2. the Require header field with value "norefersub" as specified in IETF RFC 4488
            // 3. the Refer-To header field containing the additional transferred session SCC AS
URI for PS to CS dual radio,
            // where the URI also includes the following header fields containing the information
related to the additional
            // transferred session:
            //        A. the Target-Dialog header field populated as specified in IETF RFC 4538,
containing the dialog identifier
            //        of an dialog in the early phase supporting session of the SC UE
            //        B. the Require header field populated with the option tag value "tdialog"
            //        C. the To header field populated as specified in IETF RFC 3261, containing
the value of the P-Asserted-
            //        Identity provided by the remote UE during the session establishment
            //        D. the From header field populated as specified in IETF RFC 3261, containing
the value of the P-Asserted-
```

```
            //        Identity provided by the SC UE during the session establishment
            //        E. the Content-Type header field with "application/sdp"
            //        F. the URI header field with the hname "body" populated with SDP describing
the media streams as negotiated
            //        in the session with the remote UE and
            //        G. optionally the P-Asserted-Identity URI header field containing value of
the P-Asserted-Identity header field
            //        of the received SIP INVITE request and
            // 4. application/vnd.3gpp.state-and-event-info+xml MIME body populated as follows:
            //        A if a SIP 180 Ringing response to the SIP INVITE request has already been
received in any of the early
            //        dialogs associated with the originating early session not accepted yet, with
the state-info XML element
            //        containing "early" and the direction XML element containing "initiator" and
            //        B if a SIP 180 Ringing response to the SIP INVITE request has not been
received yet in any of the early
            //        dialogs associated with the originating early session not accepted yet, with
the state-info XML element
            //         containing "pre-alerting" and the direction XML element containing
"initiator".
ensure that {
    when {
        // TODO: Trigger unclear for this refer
        the IUT receives an ACK
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an REFER containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        ReferSub indicating value false,              // 1. the Refer-Sub header field with value
"false"
        Require indicating value "norefersub",    // 2. the Require header field with value
"norefersub"
        ReferTo containing "a SIP URI with the Target-Dialog_URI_header_field"  // 3. the Refer-To
header field containing the additional transferred session SCC AS URI for PS to CS dual radio,
where the URI also includes the following header fields containing the information related to the
additional transferred session:
            TargetDialog indicating value "TODO: dialog identifier of an dialog in the early phase
supporting session of the SC UE", // A. the Target-Dialog header field populated, containing the
dialog identifier of an dialog in the early phase supporting session of the SC UE
            Require indicating value "tdialog", // B. the Require header field populated with the
option tag value "tdialog"
            To indicating value "", // C. the To header field populated, containing the value of
the P-Asserted-Identity provided by the remote UE during the session establishment
            From indicating value "", // D. the From header field populated, containing the value
of the P-Asserted-Identity provided by the SC UE during the session establishment
            ContentType indicating value "application/sdp", // E. the Content-Type header field
with "application/sdp"
            MessageBody containing  // F. the URI header field with the hname "body" populated
with SDP describing the media streams as negotiated in the session with the remote UE
                SDP containing
                    TODO indicating value ""


            ,
            PAssertedIdentity indicating value "" // G. optionally the P-Asserted-Identity URI
header field containing value of the P-Asserted-Identity header field of the received SIP INVITE
request
    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_PCT_REF_02
Test Objective	Verify that the SCC AS terminates the call over CS after receiving a SIP 488.
Reference	ETSI TS 124 237 [1], Clauses 9.3.5.5 ETSI TS 124 292 [2], Clause 10.4.7
PICS Selection	NONE
Initial Conditions	

```
// TODO with {
the UE isRegisteredTo the IMS
// TODO:
// SIP INVITE was sent with:
    //  creating the session in the terminating alerting phase without an SDP MIME body
```

```
// SCC AS supports PS to CS dual radio access
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an r_488_NotAcceptableHere containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an REFER containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        ReferSub indicating value false,              // 1. the Refer-Sub header field with value
"false"
        Require indicating value "norefersub",    // 2. the Require header field with value
"norefersub"
        ReferTo containing "a SIP URI with the Target-Dialog_URI_header_field"  // 3. the Refer-To
header field containing the additional transferred session SCC AS URI for PS to CS dual radio,
where the URI also includes the following header fields containing the information related to the
additional transferred session:
            TargetDialog indicating value "TODO: dialog identifier of an dialog in the early phase
supporting session of the SC UE", // A. the Target-Dialog header field populated, containing the
dialog identifier of an dialog in the early phase supporting session of the SC UE
            Require indicating value "tdialog", // B. the Require header field populated with the
option tag value "tdialog"
            To indicating value "", // C. the To header field populated, containing the value of
the P-Asserted-Identity provided by the remote UE during the session establishment
            From indicating value "", // D. the From header field populated, containing the value
of the P-Asserted-Identity provided by the SC UE during the session establishment
            ContentType indicating value "application/sdp", // E. the Content-Type header field
with "application/sdp"
            MessageBody containing  // F. the URI header field with the hname "body" populated
with SDP describing the media streams as negotiated in the session with the remote UE
                SDP containing
                    TODO indicating value ""


            ,
            PAssertedIdentity indicating value "" // G. optionally the P-Asserted-Identity URI
header field containing value of the P-Asserted-Identity header field of the received SIP INVITE
request
    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_PCT_RES_01
Test Objective	Verify that the SCC AS sends a SIP PRACK and SIP UPDATE after receiving a SIP 1xx.
Reference	ETSI TS 124 237 [1], Clause 9.3.5.5
PICS Selection	NONE

Initial Conditions

```
// TODO with {
the UE isRegisteredTo the IMS
// TODO:
// SIP INVITE was sent with:
    //  creating the session in the terminating alerting phase without an SDP MIME body
// SCC AS supports PS to CS dual radio access
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an r_1xx_Any containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    from the IMS_S_CSCF entity
}
```

```
  then {
        the IUT sends an PRACK containing // send a SIP PRACK request towards the CS domain
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
        and
        the IUT sends an UPDATE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        Contact indicating value "set to the Contact header field provided by the served UE at the
creation of the dialog with the remote UE", // the Contact header field set to the Contact header
field provided by the served UE at the creation of the dialog with the remote UE
        MessageBody containing  // anew SDP offer, including the media characteristics as received
in the SIP 1xx response with the SDP answer, by following the rules of 3GPP TS 24.229.
                SDP containing
                    Media indicating value "the same media characteristics as received in the SIP
1xx response with the SDP answer"


    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_MED_BYE_01
Test Objective	Verify that the SCC AS sends a SIP BYE after transfer of a session is successfully completed and the source access leg is an ongoing session containing only an active or inactive media component.
Reference	ETSI TS 124 237 [1], Clause 9.3.6
PICS Selection	NONE
Initial Conditions	

```
// TODO with {
the UE isRegisteredTo the IMS
// TODO
// source access leg is an ongoing session containing only an active or inactive media component
           }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an ACK
        // TODO: trigger for event unclear, transfer of a session is successfully completed
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an BYE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA
    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_MED_RES_01
Test Objective	Verify that the SCC AS sends a SIP 480 after transfer of a session is successfully completed and the session is dialog in an early dialog phase on the originating side.
Reference	ETSI TS 124 237 [1], Clause 9.3.6
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
// TODO: session is dialog in an early dialog phase on the originating side
           }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an ACK
        // TODO: trigger for event unclear, transfer of a session is successfully completed
    from the IMS_S_CSCF entity
```

```
}
  then {
        the IUT sends an r_480_TemporaryUnavailable containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE
    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_MED_RIN_01
Test Objective	Verify that SCC AS sends a reINVITE after transfer of a session is successfully completed and the source access leg contains media components other than speech media component.
Reference	ETSI TS 124 237 [1], Clause 9.3.6
PICS Selection	NONE
Initial Conditions	

```
// TODO with {
the UE isRegisteredTo the IMS
// TODO:
// transfer of a session is successfully completed
// and
// the source access leg contains media components other than speech media component
           }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an ACK
        // TODO: trigger for event unclear, transfer of a session is successfully completed
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an INVITE containing // reINVITE
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        ContentType indicating value "application/sdp",
        ContentLength,
        MessageBody containing
            SDP containing
                Media indicating value "TODO: the source access leg contains media components
other than speech media component"
    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_05
Test Objective	Verify that the SCC AS performs the procedures correctly for CS to PS dual radio access transfer for originating calls in pre-alerting phase or in alerting phase on the originating side.
Reference	ETSI TS 124 237 [1], Clause 9.3.7.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
// the SCC AS supports CS to PS dual radio access transfer for originating calls in pre-alerting
phase
           }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        Contact containing
            g3gppDrvccOrigPreAlertingFeature Tag,     // a g.3gpp.drvcc-orig-pre-alerting media
feature tag as described in annex C in the Contact header field was
                                                      // included in the SIP INVITE due to static
```

```
STI
        MessageBody containing
            SDP containing
                Media indicating value PX_PSMedia
    from the IMS_S_CSCF entity
}
  then {
        the IUT sends an UPDATE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        Contact indicating value "set to the Contact header field provided on the source leg at
the creation of the dialog",// a the Contact header field set to the Contact header field provided
on the source leg at the creation of the dialog with the remote UE
        ContentType indicating value "application/sdp",
            MessageBody containing  // 3 a new SDP offer, including:
                SDP containing
                    Media indicating value PX_PSMedia // TODO: PIX for media, media
characteristics as received in the SIP INVITE request


    to the IMS_S_CSCF entity
    }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_06
Test Objective	Verify that SCC AS correctly recognizes and initiated for CS to PS dual radio access transfer for a call in the alerting phase on the terminating side.
Reference	ETSI TS 124 237 [1], Clause 9.3.7.3
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS and
 the IUT hasAchievedInitialINVITE // TODO: INVITE contains: g3gppDrvccAlerting
// SCC AS supports CS to PS dual radio access transfer for calls in alerting phase
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an r_180_Ringing
        from the IMS_S_CSCF entity
}
  then {
        the IUT sends an UPDATE containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        Contact indicating value "set to the Contact header field provided on the source leg at
the creation of the dialog",// a the Contact header field set to the Contact header field provided
on the source leg at the creation of the dialog with the remote UE
        ContentType indicating value "application/sdp",
            MessageBody containing  // 3 a new SDP offer, including:
                SDP containing
                    Media indicating value PX_PSMedia // TODO: PIX for media, media
characteristics as received in the SIP INVITE request


    to the IMS_S_CSCF entity
}
                 }
```

TP Id	TP_ISC_SCCAS_PCT_INV_04
Test Objective	Verify that SCC AS correctly recognizes and initiated procedures for PS to CS dual radio access transfer of an additional session in an early dialog phase.
Reference	ETSI TS 124 237 [1], Clause 9.3.7.4
PICS Selection	NONE
Initial Conditions	

```
// PICS_SCCAS_CS2PS_DUAL_RADIO_ACCES_TRANSFER_FOR_CALLS_IN_EARLY_PHASE with {
the UE isRegisteredTo the IMS and
```

```
 the IUT hasAchievedInitialINVITE
// SCC AS the SCC AS supporting CS to PS dual radio access transfer for calls in an early phase
                }
```

Expected Behaviour

```
ensure that {
    when {
       the IUT receives an r_180_Ringing
       from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an REFER containing
        From indicating value PX_UE_A_SIP_URI,
        To indicating value PX_UE_B_SIP_URI,
        CallId indicating value PX_UE_A_CALLID,
        Via indicating value PX_UE_A_VIA,
        Route indicating value PX_UE_A_SERVICE_ROUTE,
        ReferSub indicating value false, //    1 include a Refer-Sub header field with value
"false" as specified in IETF RFC 4488
        Supported indicating value "norefersub", // 2 include a Require header field with value
"norefersub" as specified in IETF RFC 4488
        ReferTo containing  // 3 include a Refer-To header field
            TargetDialog indicating value "Target-Dialog header field containing the dialog
identifier of the dialog in the early phase, if exist", // a if an early dialog supporting the
additional transferred session exists, the Target-Dialog header field
            Require indicating value "tdialog", // c the Require header field populated with the
option tag value "tdialog"
            From indicating value "value of the P-Asserted-Identity provided by the SC UE during
the session establishment", //the From header field populated as specified in IETF RFC 3261,
containing the value of the P-Asserted-Identity provided by the SC UE during the session
establishment
            ContentType indicating value "application/sdp", // E. the Content-Type header field
with "application/sdp"
            MessageBody containing  // F. the URI header field with the hname "body" populated
with SDP describing the media streams as negotiated in the session with the remote UE
                SDP containing
                    TODO indicating value ""
                ,
        Accept containing  // if a SIP 180 Ringing response to the INVITE request has already been
received on the source access leg, the state-info XML element containing "early" and the direction
XML element containing "receiver"
            MIME indicating value "application/vnd.3gpp.state-and-event-info.xml" //subclause
D.2.3
        to the IMS_S_CSCF entity
     }
}
```

5.2.2.7 PS-PS access transfer

TP Id	TP_ISC_SCCAS_PPT_RIN_01
Test Objective	Verify that the SCC AS successfully processes an INVITE on a previously established dialog if the SIP INVITE request contains a Replaces header field.
Reference	ETSI TS 124 237 [1], Clause 10.3.2
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasEstablishedDialog
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an INVITE containing // reINVITE
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
```

```
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI // added by SCC AS
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_PPT_RIN_02
Test Objective	Verify that the SCC AS successfully processes an UPDATE on an existing early dialog if the SIP INVITE request contains a Replaces header field.
Reference	ETSI TS 124 237 [1], Clause 10.3.2
PICS Selection	NONE
Initial Conditions	

```
//PICS_SCCAS_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasInitiatedDialogWith
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an UPDATE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Contact indicating value PX_UE_A_SIP_URI // added by SCC AS
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_PPT_RIN_03
Test Objective	Verify that the SCC AS processes an INVITE if it is unable to associate the SIP INVITE request with a confirmed dialog or on a previously established dialog.
Reference	ETSI TS 124 237 [1], Clause 10.3.2
PICS Selection	NONE
Initial Conditions	

```
// PICS_SCCAS_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and  // include g.3gpp.pstops-sti
the UE not hasEstablishedDialog and
the UE not hasInitiatedDialogWith
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an r_480 containing // reject Temporarily Unavailable
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID
         to the IMS_S_CSCF entity
        }
    }
```

TP Id	TP_ISC_SCCAS_PPT_RIN_04
Test Objective	Verify that the SCC AS processes an INVITE if number of media lines in the Target Access Leg is less than the number of media lines in the Source Access Leg.
Reference	ETSI TS 124 237 [1], Clause 10.3.2
PICS Selection	NONE
Initial Conditions	

// PICS_SCCAS_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasEstablishedDialog
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
TargetDialog indicating value PX_UE_MediaLines // indicating number of media lines in the Target Access Leg is less than the number of media lines in the Source Access Leg
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_480 containing // reject Temporarily Unavailable
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_PPT_BYE_01
Test Objective	Verify that the SCC AS successfully processes an BYE on a previously established dialog, if does not receive within this time interval an initial SIP INVITE request.
Reference	ETSI TS 124 237 [1], Clause 10.3.4
PICS Selection	NONE
Initial Conditions	

// PICS_SCCAS_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasEstablishedDialog
}	

Expected Behaviour	

ensure that {
when {
. at time point t1: the IUT receives an BYE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
Reason indicating value 480 // status code 480 Temporarily Unavailable
from the IMS_S_CSCF entity
}
then {
! "after time interval of 8 seconds" after t1: the IUT sends an r_200 containing // 200 OK BYE to UE_B
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
and
the IUT sends an BYE containing // BYE to UE_A
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_PPT_BYE_02
Test Objective	Verify that the SCC AS successfully processes an CANCEL on a early dialog.
Reference	ETSI TS 124 237 [1], Clause 10.3.6
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_RELEASING_EARLY_DIALOG_DURING_PS2PS_ACCES_TRANSFER with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasInitiatedDialogWith
}	

Expected Behaviour	

ensure that {
when {
. at time point t1: the IUT receives an BYE containing // status code 503
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE
from the IMS_S_CSCF entity
}
then {
! "after time interval of 8 seconds" after t1: the IUT sends an r_200 containing // 200 OK BYE
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
and
the IUT sends an CANCEL containing // CANCEL to UE_B
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_PPT_CAN_01
Test Objective	Verify that the SCC AS successfully processes an CANCEL on a early dialog.
Reference	ETSI TS 124 237 [1], Clause 10.3.6
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasInitiatedDialogWith
}	

Expected Behaviour	

ensure that {
when {
. at time point t1: the IUT receives an CANCEL containing // status code 503
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE
from the IMS_S_CSCF entity
}
then {
! "after time interval of 8 seconds" after t1: the IUT sends an r_200 containing // 200 OK CANCEL
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_PPT_RES_01
Test Objective	Verify that the SCC AS successfully processes an SIP 503 (Service Unavailable) response on a early dialog.
Reference	ETSI TS 124 237 [1], Clause 10.3.6
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and // include g.3gpp.pstops-sti
the UE hasInitiatedDialogWith
}	

Expected Behaviour	

ensure that {
when {
. at time point t1: the IUT receives an r_503 containing // from UE_B
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE
from the IMS_S_CSCF entity
}
then {
! "after time interval of 8 seconds" after t1: the IUT sends an r_503 containing // to UE_A
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
to the IMS_S_CSCF entity
}
}	

5.2.2.8 PS-PS access transfer in conjunction with PS-CS access transfer

TP Id	TP_ISC_SCCAS_CON_RIN_01
Test Objective	Verify that the SCC AS successfully processes an INVITE that includes an active speech media component using CS bearer.
Reference	ETSI TS 124 237 [1], Clause 11.3.2
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2PS_ACCES_TRANSFER_IN_CONJUNCTION_PS2CS with {
the UE isRegisteredTo the IMS and
the UE hasEstablishedDialog
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing // request active speech media component using CS bearer
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
from the IMS_S_CSCF entity
}
then {
the IUT sends an INVITE containing // reINVITE
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Contact indicating value PX_UE_A_SIP_URI // added by SCC AS
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CON_RIN_02
Test Objective	Verify that the SCC AS successfully processes an INVITE due to two STIs on the Target Access Leg.
Reference	ETSI TS 124 237 [1], Clause 11.3.3
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2PS_ACCES_TRANSFER_IN_CONJUNCTION_PS2CS with {
the UE isRegisteredTo the IMS and
the UE hasEstablishedDialog
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing // request CS -> PS
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
//Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
from the IMS_S_CSCF entity
}
then {
the IUT sends an INVITE containing // reINVITE
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
//Contact indicating value PX_UE_A_SIP_URI // added by SCC AS
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CON_RIN_03
Test Objective	Verify that the SCC AS rejects an INVITE due missing ability to associate it to an ongoing or different SIP dialog.
Reference	ETSI TS 124 237 [1], Clause 11.3.3
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2PS_ACCES_TRANSFER_IN_CONJUNCTION_PS2CS with {
the UE isRegisteredTo the IMS and
the UE hasEstablishedDialog
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing // request CS -> PS
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
//Replaces indicating value PX_UE_A_SIP_URI // indicating access transfer procedure
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_480 containing // reINVITE
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
//Contact indicating value PX_UE_A_SIP_URI // added by SCC AS
to the IMS_S_CSCF entity
}
}	

5.2.2.9 PS-CS access transfer, Single Radio

TP Id	TP_ISC_SCCAS_SRA_INV_01
Test Objective	Verify that the SCCAS distinguish SIP INVITE with STN-SR for nonexisting dialog.
Reference	ETSI TS 124 237 [1], Clause 12.3.1
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a INVITE containing
RequestUri indicating value PX_STI_URI,
From indicating value PX__SIP_URI,
To indicating value PX__SIP_URI,
CallId indicating value PX_STI_CALLID,
Via indicating value PX_STI_VIA,
Contact,
TargetDialog containing "dialog info from nonexisting PS session",
Require indicating value "tdialog",
PAssertedIdentity indicating value PX_UE_CMSISDN
SipMessageBody
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_480_TemporaryUnavailable
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_SRA_INV_02
Test Objective	Verify that the SCCAS distinguish SIP INVITE with STN-SR and informs remote UE.
Reference	ETSI TS 124 237 [1], Clause 12.3.1
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a INVITE containing
RequestUri indicating value PX_STNSR_URI,
From indicating value PX__SIP_URI,
To indicating value PX__SIP_URI,
CallId indicating value PX_STI_CALLID,
Via indicating value PX_STNSR_VIA,
Contact,
TargetDialog containing "dialog info from PS session",
Require indicating value "tdialog",
PAssertedIdentity indicating value PX_UE_CMSISDN
SipMessageBody
from the IMS_S_CSCF entity
}
then {
the IUT sends a ReINVITE //for remote UE leg
to the IMS_S_CSCF and
the IUT receives a r_200_Ok and
the IUT sends a r_200_Ok
}
}	

TP Id	TP_ISC_SCCAS_SRA_INV_03
Test Objective	Verify that the SCCAS distinguish Transfer of first session.
Reference	ETSI TS 124 237 [1], Clauses 12.3.2.1 and 12.3.2.2
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
        }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INVITE containing
            RequestUri indicating value PX_STI_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STI_VIA,
            Contact containing
                g3gppMidcallMediaFeature Tag
            TargetDialog containing "dialog info existing PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody
        from the IMS_S_CSCF and
        the IUT receives a  r_200_Ok
    }
    then {
        the IUT sends a r_200_Ok
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_SRA_INV_04
Test Objective	Verify that the SCCAS distinguish SIP INVITE with STN-SR and not release source leg.
Reference	ETSI TS 124 237 [1], Clause 12.3.1
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a ReINVITE containing
            RequestUri indicating value PX_STNSR_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STNSR_VIA,
            Contact,
            TargetDialog containing "dialog info from PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody,
            Reason containing Cause indicating value "487"

        from the IMS_S_CSCF entity
    }
    then {
        the IUT not sends a BYE //for source UE leg
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_SRA_BYE_01
Test Objective	Verify that the SCCAS distinguish SIP INVITE with STN-SR and not release source leg.
Reference	ETSI TS 124 237 [1], Clause 12.3.3.2
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a BYE containing
From indicating value PX_IMS_S_CSCF_SIP_URI,
To indicating value PX_UE_A_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Reason containing Cause indicating value "503"
 	
from the IMS_S_CSCF entity
}
then {
the IUT receives a INVITE and
the IUT not sends a BYE //forÂ¸ source UE leg
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_SRA_BYE_02
Test Objective	Verify that the SCCAS distinguish SIP BYE with STN-SR and release source leg after OperatiorSpecificTimer timeout.
Reference	ETSI TS 124 237 [1], Clause 12.3.3.4
PICS Selection	NONE
Initial Conditions	

//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM and
the UE hasEstablishedPS2CSDialog
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a BYE containing
RequestUri indicating value PX_STNSR_URI,
From indicating value PX__SIP_URI,
To indicating value PX__SIP_URI,
CallId indicating value PX_STI_CALLID,
Via indicating value PX_STNSR_VIA,
Contact,
TargetDialog containing "dialog info from PS session",
Require indicating value "tdialog",
PAssertedIdentity indicating value PX_UE_CMSISDN
SipMessageBody,
Reason containing Cause indicating value "Reason Q.850 31"
 	
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_200_Ok_BYE and
the IUT on timerOperatorSpecificTimeout sends a BYE //for source UE leg
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_SRA_BYE_03
Test Objective	Verify that the SCCAS distinguish SIP BYE with STN-SR and release source leg on BYE Cause different from Cause Q850 31.
Reference	ETSI TS 124 237 [1], Clause 12.3.3.4
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasEstablishedDialog the SCCAS_UE_TERM and
the UE hasEstablishedPS2CSDialog
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a BYE containing
            RequestUri indicating value PX_STNSR_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STNSR_VIA,
            Contact,
            TargetDialog containing "dialog info from PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody,
            Reason containing Cause not indicating value "Reason Q.850 31"

        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok_BYE and
        the IUT sends a BYE //for source UE leg
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_SRA_CAN_01
Test Objective	Verify that the SCCAS distinguish SIP CANCEL and release session transfered leg after OperatiorSpecificTimer timeout.
Reference	ETSI TS 124 237 [1], Clause 12.3.3.5.1, 1A
PICS Selection	NONE

Initial Conditions

```
//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasInitiatedDialogWith the SCCAS_UE_TERM and
the UE hasInitiatedPS2CSinEarlydialog
               }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a CANCEL containing
            RequestUri indicating value PX_STNSR_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STNSR_VIA,
            Contact,
            TargetDialog containing "dialog info from PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody,
            Reason containing Cause not indicating value "Reason Q.850 31"

        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok_CANCEL and
        the IUT on timerOperatorSpecificTimeout sends a CANCEL
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_SRA_CAN_02
Test Objective	Verify that the SCCAS send 200Ok_CANCEL and release PStoCS session transfered leg in alerting state for terminating user.
Reference	ETSI TS 124 237 [1], Clause 12.3.3.5.2, 1A
PICS Selection	NONE
Initial Conditions	

```
//PICS_SCCAS_PS2CS_SRVCC with {
the UE isRegisteredTo the IMS_A and
the UE hasInitiatedDialogWith the SCCAS_UE_TERM and
the UE hasInitiatedPS2CSinAlertingState
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a CANCEL containing
            RequestUri indicating value PX_STNSR_URI,
            From indicating value PX__SIP_URI,
            To indicating value PX__SIP_URI,
            CallId indicating value PX_STI_CALLID,
            Via indicating value PX_STNSR_VIA,
            Contact,
            TargetDialog containing "dialog info from PS session",
            Require indicating value "tdialog",
            PAssertedIdentity indicating value PX_UE_CMSISDN
            SipMessageBody,
            Reason containing Cause not indicating value "Reason Q.850 31"

        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok_CANCEL //and
        //the IUT on timerOperatorSpecificTimeout sends a CANCEL
        //to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_PCT_INV_06
Test Objective	Verify that SCC AS correctly recognizes and initiated for PS to CS access transfer for terminating call in pre-alerting phase using PS to CS SRVCC procedure.
Reference	ETSI TS 124 237 [1], Clause 12.3.4.2
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
                }
```

Expected Behaviour

```
ensure that {
    when {
       the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            Contact containing
                g3gpppps2csSrvccTermPreAlerting
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_183_SessionProgress
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_PCT_INV_02
Test Objective	Verify that SCC AS correctly recognizes and initiated for PS to CS access transfer for originating call in pre-alerting phase using PS to CS SRVCC procedure.
Reference	ETSI TS 124 237 [1], Clause 12.3.4.3
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
Contact containing
g3gppPs2csSrvccOrigPreAlerting
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_183_SessionProgress // possible r_18x
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_PCT_INV_03
Test Objective	Verify that SCC AS correctly procedures for PS to CS access transfer of additional call.
Reference	ETSI TS 124 237 [1], Clause 12.3.4.4
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE // TODO: INVITE contains: g3gppSrvccAlerting
// TODO: and more conditions to distinguish subchapter from 12.3.4.2, 12.3.4.3
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an r_180_Ringing
from the IMS_S_CSCF entity
}
then {
the IUT sends an REFER containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
ReferSub indicating value false,
Supported indicating value "norefersub"
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_MED_OPT_01
Test Objective	Verify that the SCC AS handles correctly the procedures for handling of SIP OPTIONS request.
Reference	ETSI TS 124 237 [1], Clause 12.3.7
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an OPTIONS containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,	

```
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            ReferTo containing
                   PAssertedIdentity indicating value "the C-MSISDN in the P-Asserted-Identity
header field that the session that was most recently made"
         from the IMS_S_CSCF entity
    }
    then {
        the IUT sends a r_200_Ok containing
        ContentType indicating value "application/sdp",
           ContentLength,
        MessageBody containing
            SDP containing
                Media indicating valuee "audio and video"
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_07
Test Objective	Verify that the SCC AS handles correctly the transfer of session with active speech media component.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.2
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
TargetDialog
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_200_Ok //
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CPT_INV_08
Test Objective	Verify that the SCC AS handles correctly the transfer of session with inactive speech media component.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.3
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
TargetDialog
from the IMS_S_CSCF entity
}
then {
the IUT sends an INVITE containing // reINVITE
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA
to the IMS_S_CSCF entity	

```
        }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_09
Test Objective	Verify that the SCC AS handles correctly the transfer of originating session in alerting phase.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.2.4
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE and // INVITE initiated by served User and contains TargetDialog header field
the IUT hasAchievedUPDATEuponINVITE
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an r_200_Ok
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_180_Ringing
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CPT_INV_10
Test Objective	Verify that the SCC AS handles correctly the transfer of terminating session in alerting phase.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.2.5
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE and // INVITE initiated by remote user and contains TargetDialog header field
the IUT hasAchievedUPDATEuponINVITE
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an r_200_Ok
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_183_SessionProgress
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_CPT_INV_11
Test Objective	Verify that the SCC AS handles correctly additional session transfer initiation.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.3.2
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an INVITE containing
From indicating value PX_UE_A_SIP_URI,
To indicating value PX_UE_B_SIP_URI,
CallId indicating value PX_UE_A_CALLID,
Via indicating value PX_UE_A_VIA,
Route indicating value PX_UE_A_SERVICE_ROUTE,
Accept containing
MIME indicating value "application/vnd.3gpp.mid-call+xml"
from the IMS_S_CSCF entity
}	

```
    then {
        the IUT sends a r_200_Ok
        to the IMS_S_CSCF entity
     }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_12
Test Objective	Verify that the SCC AS handles correctly transfer of session with inactive speech media component.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.3.3
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS
                }
```

Expected Behaviour

```
ensure that {
    when {
       the IUT receives an INVITE containing
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA,
            Route indicating value PX_UE_A_SERVICE_ROUTE,
            Accept containing
                MIME indicating value "application/vnd.3gpp.mid-call+xml"
        from the IMS_S_CSCF entity
    }
    then {
        the IUT sends an INVITE containing // reINVITE
            From indicating value PX_UE_A_SIP_URI,
            To indicating value PX_UE_B_SIP_URI,
            CallId indicating value PX_UE_A_CALLID,
            Via indicating value PX_UE_A_VIA
         to the IMS_S_CSCF entity
        }
}
```

TP Id	TP_ISC_SCCAS_CPT_INV_13
Test Objective	Verify that the SCC AS handles correctly transfer of originating session in alerting phase.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.3.4
PICS Selection	NONE
Initial Conditions	

```
with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE and // INVITE initiated by served User and contains Accept
containing MIME indicating value "application/vnd.3gpp.mid-call+xml"
the IUT hasAchievedUPDATEuponINVITE
                }
```

Expected Behaviour

```
ensure that {
    when {
       the IUT receives an r_200_Ok
       from the IMS_S_CSCF entity
    }
then {
    the IUT sends an r_180_Ringing
     to the IMS_S_CSCF entity
    }
                }
```

TP Id	TP_ISC_SCCAS_CPT_INV_14
Test Objective	Verify that the SCC AS handles correctly transfer of terminating session in alerting phase.
Reference	ETSI TS 124 237 [1], Clause 12.3.10.3.5
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE and // INVITE initiated by remote user and contains Accept containing MIME indicating value "application/vnd.3gpp.mid-call+xml"
the IUT hasAchievedUPDATEuponINVITE
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an r_200_Ok
from the IMS_S_CSCF entity
}
then {
the IUT sends an r_183_SessionProgress
to the IMS_S_CSCF entity
}
}	

TP Id	TP_ISC_SCCAS_TER_BYE_01
Test Objective	Verify that the SCC AS correctly behave procedures when the access transfer is completed.
Reference	ETSI TS 124 237 [1], Clause 12.3.11
PICS Selection	NONE
Initial Conditions	

with {
the UE isRegisteredTo the IMS and
the IUT hasAchievedInitialINVITE
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a BYE containing
RequestUri indicating value PX_STNSR_URI,
From indicating value PX__SIP_URI,
To indicating value PX__SIP_URI,
CallId indicating value PX_STI_CALLID,
Via indicating value PX_STNSR_VIA,
Contact,
TargetDialog containing "dialog info from PS session",
Require indicating value "tdialog",
PAssertedIdentity indicating value PX_UE_CMSISDN
SipMessageBody,
Reason containing Cause indicating value "Reason Q.850 1" // should be different then 31
 	
from the IMS_S_CSCF entity
}
then {
the IUT sends a r_200_Ok_BYE and
the IUT sends a r_404_NotFound //according to 3GPP TS 29.292 table 5.4.8.1.1 and table 5.4.8.1.2
to the IMS_S_CSCF entity
}
}	

Annex A (normative): TDL-TO source files

Each TP in clause 5.2 above has been written in TDL-TO and thus in a structured manner which is consistent with all other TPs. The TDL-TO text files for all test purposes are contained in archive ts_10383302v010101p0.zip which accompanies the present document.

History

Document history		
V1.1.1	December 2022	Publication